



Dominique Makowski

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*Trained as a clinical neuropsychologist and CBT psychotherapist, I am an assistant professor at the University of Sussex, where I head the [Reality Bending Lab](#). My research has three strands. **Reality and its distortions:** beliefs about AI-generated content, visual illusions, deception, misinformation and fake news, fiction, and altered states of consciousness. **The body in cognition:** how interoception, physiological states and emotion regulation shape what we take to be real. **Methods:** Bayesian statistics, computational models and psychophysiology, as well as the development of new software, measures and paradigms, released as open source ([easystats](#), [NeuroKit](#), [cogmod](#)) to support open and rigorous science.*

Experience



Assistant Professor (Current Position)

SCHOOL OF PSYCHOLOGY, UNIVERSITY OF SUSSEX

UK

2023 - Present

- Full teaching service on courses including Bayesian Statistics and Psychophysiology.
- Head of the [Reality Bending Lab](#): two doctoral students to date, around two full-time placement students and a dozen dissertation students a year, funded summer interns and research assistants.
- School-level leadership responsibilities, including research ethics, the research participation scheme and postgraduate programme development.



Research Fellow (Postdoc)

SCHOOL OF SOCIAL SCIENCES, NANYANG TECHNOLOGICAL UNIVERSITY (NTU)

Singapore

2019 - 22

- Awarded the NTU Presidential Postdoctoral Fellowship (2021 - 22), a competitive two-year grant bridging postdoc and faculty. Led an independent research track on the neurocognitive and affective processes underlying beliefs about reality within the [Clinical Brain Lab](#), managing a research budget and a team of three research assistants.
- Research Fellow (2019 - 20) on a Singapore Ministry of Defence funded project on the neurocognitive correlates of deception: study design, data collection, analysis and publications; co-supervision of one PhD student and two research assistants.

Education



PhD - Psychology

UNIVERSITÉ PARIS CITÉ

France

2014 - 18



Certified Psychotherapist - Cognitive Behavioural Therapy (CBT)

FRENCH ASSOCIATION FOR COGNITIVE BEHAVIOURAL THERAPY (AFTCC)

France

2014 - 17



BSc, MSc - Clinical Neuropsychology and Cognitive Psychology

UNIVERSITÉ PARIS CITÉ

France

2009 - 14

Clinical Practice



Assistant Neuropsychologist

ADULT PSYCHIATRY, HÔPITAL PITIÉ-SALPÊTRIÈRE, PARIS

France

2013 - 14

- Professional placement (580 h): neuropsychological assessment and diagnosis of inpatients with mood disorders, psychosis and attention deficit hyperactivity disorder, and cognitive remediation.



Assistant Neuropsychologist

GERIATRIC PSYCHIATRY, HÔPITAL SAINTE-ANNE, PARIS

France

2012 - 13

- Professional placement (304 h): assessment and diagnosis of complex presentations combining psychiatric disorders with early dementia in older adults.



Assistant Neuropsychologist

GERONTOLOGY, HÔPITAL LA PORTE VERTE, VERSAILLES

France

2012

- Observation placement (96 h): assessment of memory deficits in neurodegenerative disease, somatic ageing-related illness and depression.

Languages



French
Native



Polish
Native



English
Fluent



Italian
B1 · learning



R
Fluent



Python
Fluent

Teaching

I have been teaching at university level since 2013, and I am a Fellow of the Higher Education Academy (FHEA). At Sussex my involvement is primarily with advanced and postgraduate courses, but I am equally comfortable with introductory and undergraduate content.

METHODS AND STATISTICS

Bayesian Statistics	MSc · PhD	Module convenor ; designed from scratch. The Bayesian framework, estimation and inference; multilevel and cognitive models with brms. UNIVERSITY OF SUSSEX · 2024 - PRESENT
Advanced Statistics	BSc · MSc · PhD	Regression and mixed models; psychometrics; experimental methodology and scientific reasoning. Taught in R . UNIVERSITY OF SUSSEX · 2023 - PRESENT · UNIVERSITÉ PARIS CITÉ · 19 H · 2017 - 18
Programming and Tools	MRes · PhD	Online experiment building with JavaScript; signal processing with Python ; version control and collaboration with GitHub ; Open data and materials; agentic coding (AI) tools for research. UNIVERSITY OF SUSSEX · 2025 · UNIVERSITY OF ZURICH · 24 H · 2022

COGNITION AND PSYCHOBIOLOGY

Psychophysiology	BSc · MSc	Neurophysiological signal recording and preprocessing; heart rate variability; interoception; embodied cognition. UNIVERSITY OF SUSSEX · 2023 - PRESENT · UNIVERSITY OF BASEL · 8 H · 2024
Cognitive Neuroscience	BSc	Neuroanatomy; memory; emotion regulation; neuroimaging; mindfulness and meditation. UNIVERSITÉ PARIS CITÉ · 90 H · 2017 - 18
Cognitive Psychology	BSc	Models of working memory; mental imagery and categorisation. UNIVERSITÉ PARIS CITÉ · 48 H · 2014 - 16

INDIVIDUAL DIFFERENCES AND CLINICAL PSYCHOLOGY

Individual Differences	BSc	Intelligence; personality; creativity. UNIVERSITÉ PARIS CITÉ · 144 H · 2015 - 17
Clinical Neuropsychology	BSc	Neuropsychological syndromes; neuropsychological examination, diagnosis and psychometrics; neuropsychology in psychiatry. UNIVERSITÉ PARIS CITÉ · 51 H · 2017 - 18
Rehabilitation and Practice	MSc	Cognitive rehabilitation and intervention; neuropsychology and disability; neurodegenerative and neurodevelopmental disorders; clinical practice supervision. UNIVERSITÉ PARIS CITÉ · 29 H · 2017 - 18

DESIGN AND DELIVERY

MSc Programme development	2026
UNIVERSITY OF SUSSEX Lead on designing a new postgraduate programme: MSc Advanced Computational Psychology with its curriculum structure, module content and learning outcomes.	
Module design	2024
UNIVERSITY OF SUSSEX Created Bayesian Statistics as a new master's module (also attended by PhDs). Student's evaluation highlight it as one of the most engaging modules.	
Invited teaching	2021 - 25
BASEL · ZURICH · PARIS · SINGAPORE · OHBM I am regularly invited to deliver specific upskilling trainings for Staff or PhDs, with workshops on physiological techniques, programming or statistics.	

SUPERVISION AND EXAMINING

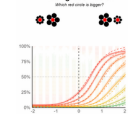
Doctoral supervision and examining	4 examined
Two PhD students supervised at Sussex, one graduated. Internal examiner for four Sussex PhD theses, on consciousness science, interoception and quantitative methods.	
Dissertation and placement students	12 a year
Supervising nine undergraduate and three master's dissertations a year, two full-time placement students, and funded summer interns.	
Student prizes and scholarships	8 since 2023
Supervised students have won a SAGE prize for outstanding application of statistical methods , a Sussex Partnership NHS Trust prize for the highest project mark, an Adam Weiler PhD Impact Award, and five JRA and SoCoBio summer scholarships.	

Software and Tools

SOFTWARE

	easystats R · CO-FOUNDER · 2019 - PRESENT Ten R packages for the post-processing of statistical models: parameters, effect sizes, diagnostics, Bayesian indices, marginal effects, reporting. Used in over 10,000 publications . SIPS Mission Award (2023).	41.2M downloads
	NeuroKit PYTHON · CREATOR · 2017 - PRESENT Most popular Python package for signal processing (ECG, PPG, EDA, ...), from peak detection to heart rate variability and complexity indices. 2,355 stars and dozens of contributors. SIPS Commendation (2024).	130k downloads/month
	Pyllusion PYTHON · CREATOR · 2020 - PRESENT Generates visual illusions parametrically, separating illusion strength from the objective difference between the targets, so that sensitivity can be measured continuously.	82 stars
	cogmod R · CREATOR · 2025 - PRESENT Bayesian models of cognitive tasks - drift diffusion, linear ballistic accumulator, racing diffusion - as custom brms families.	22 stars

MEASURES AND PARADIGMS

	Illusion Game SCIENTIFIC REPORTS · 2023 Parametric task measuring visual illusion sensitivity as a continuous trait, with evidence for a general factor.
	LIE Scale CURRENT PSYCHOLOGY · 2021 16-item measure of lying styles and attitudes: ability, frequency, negativity and contextuality.
	MINT Scale PSYARXIV PREPRINT Multimodal interoception questionnaire, across cardiac, respiratory, gastric and other bodily channels.

Service and Leadership

SCHOOL AND UNIVERSITY

- Programme Developer

2026 - Present

MSC ADVANCED COMPUTATIONAL PSYCHOLOGY · UNIVERSITY OF SUSSEX
- Academic Lead

2025 - Present

RESEARCH PARTICIPATION SCHEME · SCHOOL OF PSYCHOLOGY, SUSSEX
Redesigned the credit allocation and study approval rules to be fairer and more transparent, and trialled changes that raise the educational value of taking part while improving data quality.
- Seminar Organiser

2024 - Present

MORSE · CENTRE FOR OPEN SCIENCE AND RESEARCH REFORM, SUSSEX
Methods for Open and Reproducible Science: a seminar series on openness and reproducibility in psychological science. Member of the Centre.
- Committee Member

2023 - Present

CROSS-SCHOOLS RESEARCH ETHICS COMMITTEE · UNIVERSITY OF SUSSEX

Grants and Awards

GRANTS

- Measuring the Deep Self

€58,843

BIAL FOUNDATION · GRANTS PROGRAMME 2026/27 · UNDER REVIEW
Prior-weighting in predictive processing as a single trait across belief updating, perception and self-reference.
- Illusion Sensitivity and Phenomenological Control

£20,000

CENTRE FOR OPEN SCIENCE · REGISTERED REPORTS IN CONSCIOUSNESS RESEARCH · PI · 2024 - 26
- Perception of Real vs. Fake Information

£655,000

TEMASEK LABORATORIES, NTU · BEHAVIOURAL AND COMMUNICATION SCIENCE PROGRAMME · CO-PI · 2022 - 25
- Crowdedness and Environment Typology (NICE)

£179,000

SINGAPORE NATIONAL RESEARCH FOUNDATION · INTRA-CREATE GRANT · CO-PI · 2021 - 22
- “That’s Fake News”: Belief in Deceptive Information

£258,000

NANYANG TECHNOLOGICAL UNIVERSITY · PRESIDENTIAL POSTDOCTORAL FELLOWSHIP · PI · 2020 - 22

EDITORIAL AND PEER REVIEW

- Statistical Editor

2025 - Present

ROYAL SOCIETY OPEN SCIENCE
- Associate Editor

2020 - 24

FRONTIERS IN PSYCHOLOGY · QUANTITATIVE PSYCHOLOGY AND MEASUREMENT
- Regular Reviewer

2018 - Present

JOURNAL OF OPEN SOURCE SOFTWARE
- Reviewer

2017 - Present

NATURE HUMAN BEHAVIOUR · BIOLOGICAL PSYCHIATRY · SCIENTIFIC REPORTS
Also Acta Psychologica, Journal of Experimental Psychology: Human Perception and Performance, Neuropsychologia, Personality and Individual Differences and others.

AWARDS

- SIPS Commendation

2024

SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE
For the NeuroKit Python package.
- Openness in Research Award

2024

UNIVERSITY OF SUSSEX RESEARCH CULTURE AWARDS
For building software that makes open research accessible, and for running fully transparent studies.
- School of Psychology Award

2024

UNIVERSITY OF SUSSEX
For outstanding student supervision.
- Mission Award

2023

SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE
For the easystats collection of R packages.
- Best Article Award

2020

PSYCHONOMIC SOCIETY
For the fictional reappraisal paper, the main output of the PhD.
- SIPS Commendation

2020

SOCIETY FOR THE IMPROVEMENT OF PSYCHOLOGICAL SCIENCE
For the bayestestR R package.

Engagement and Knowledge Exchange

PUBLIC ENGAGEMENT AND OUTREACH

- Open Science Champion

2024

BRITISH NEUROSCIENCE ASSOCIATION · MEMBERS’ MEETING
Invited to argue the open science side of the meeting’s debate - whether neuroscience needs a **sustainability** revolution more than an **openness** one - against the Association’s Sustainability Champion and a representative of Wellcome.
- Popular Writing and Media

40+ posts

REALITY BENDING LAB · TELEVISION · PODCAST · 2015 - PRESENT
Methods tutorials, research explainers and essays on research culture, for a general and early-career audience. French national television on violent images and the brain; the *Learning Bayesian Statistics* podcast; Pint of Science; a public debate on sensory illusions in Bordeaux; press coverage of the 3D-cinema memory and AI-generated faces studies.

KNOWLEDGE EXCHANGE AND CONSULTANCY

- Scientific Advisor

2025 -

ARTIFICIAL INTELLIGENCE START-UP, SINGAPORE · CLIENT CONFIDENTIAL
Scientific advising on an applied project on deception detection.
- Contract Research

2026

FRAGRANCE INDUSTRY · CLIENT CONFIDENTIAL
Contract research on the relationship between smells and emotions.
- Contracted Instructor

2026

SOUTH EAST DOCTORAL TRAINING ARC (SEDARC)
Reusable *Introduction to GitHub* lecture for PhD students across the six universities of the partnership.
- Contracted Developer

2025

INSTITUTE FOR ADVANCED CONSCIOUSNESS STUDIES
Hyperscanning capabilities for open-source portable electroencephalography (EEG) software.

SCIENCE ADJACENT PROJECTS



NeuropsychXart

OHBM 2023

BRAIN ART EXHIBITION, MONTREAL · SELECTED
A person’s heart or brain activity, recorded during a mental experience and drawn as the **attractor** of that signal: a portrait made of physiology, where the artist is only the facilitator.



Reality Check!

In testing

CARD GAME · BETA-TESTED IN THE LAB, 2024
A bluffing card game built on the **Ebbinghaus illusion**: play a circle the same size or bigger than the last, and call the bluff when your eyes disagree. Out of the lab’s work on illusion sensitivity.

Talks and Presentations

INVITED TALKS AND SEMINARS

Reality as an Embodied Process
NANYANG TECHNOLOGICAL UNIVERSITY, SINGAPORE
Departmental research seminar on the role of the body in higher cognition, and on the tools - measures, paradigms and software - built to study it.

2025

Good Tools Breed Good Science
BRAINHACK · SINGAPORE
On how the available software, hardware and methods quietly decide which questions a field is able to ask.

2026

easystats and NeuroKit
OHBM · SUSSEX · BRIGHTON · 2023 - 2026
Invited software talks and demonstrations, including *Get Sh*t Done in R with easystats* at EARL Brighton, NeuroKit at an OHBM satellite event on research software, and the past, present and future of easystats for the Centre for Open Science and Research Reform.

4 talks

The Lying Brain
TEMASEK LABORATORIES · SINGAPORE
On the neuroscientific approaches to deception, and the case for treating it as a multidimensional phenomenon rather than a signal to be detected.

2021

CONFERENCE PRESENTATIONS

Symposia Chaired
AISC, ROME · ESCAN, ROME
The Value of Reality in the Post-Truth Era and Artificial Emotion: two international symposia on how beliefs about artificiality shape affective and neural responses, convened with collaborators in Rome.

2024 - 26

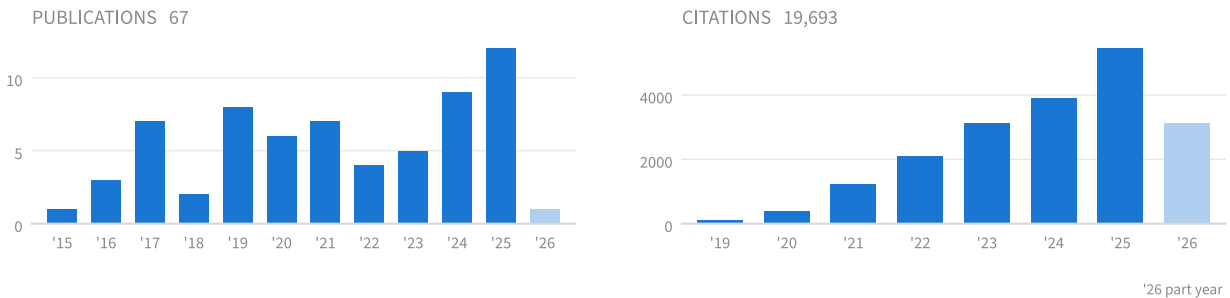
Conference Talks
ESCOP · ICOM · ICM · SFP · AFTCC · MBB · 2013 - 2023
Talks at European and international meetings in cognitive psychology, consciousness science, neuroimaging and clinical psychology, in English and in French.

13

Posters
OHBM · SNLF · SOCIÉTÉ DES NEUROSCIENCES · 2013 - 2023
Including an EEG-attractor artwork, *Unicity through Complexity*, shown in the Neuroscience x Art track at OHBM 2023.

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Research Impact



- Most cited researcher in the School of Psychology, University of Sussex, since 2024.
 - *h*-index 27, *i10*-index 38; 40 of 67 publications as first (29) or senior (11) author.
 - Concepts and tools from this work are now documented independently of their authors: the [probability of direction](#), [easystats](#) and [NeuroKit](#).
- GOOGLE SCHOLAR, JUNE 2026